Cross-Lingual Correctness Probe Transfer: Negative Results, Methodological Warnings, and Conditional Solutions

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Abstract

Probe-based calibration methods extract internal correctness directions from LLM representations, but all published work is English-only. We present the first large-scale study of cross-lingual correctness probe transfer, testing three architectures (Qwen3-235B MoE, Llama 4 Scout MoE, Gemma 4 31B Dense) across 20 typologically diverse languages. Our findings are predominantly negative but informative. (1) **Control task warning**: extending Hewitt & Liang selectivity checks to cross-lingual probing reveals that 35% of language–layer combinations are capacity artifacts (LOW-resource: 42%, MID: 31%), with selectivity strongly correlated with zero-shot AUROC (ρ=0.65). This implies many published positive cross-lingual probing results may reflect artifacts rather than genuine signal transfer. (2) **Mean-shift impossibility**: under StandardScaler + logistic regression, mean-shift produces ΔAUROC $\equiv 0$ by rank-invariance, confirmed across all 60 model–language pairs. (3) **Distributional matching path closed**: CaCLR (MMD), CORAL, and high-dimensional MMD (PCA d=50, 100) all degrade performance; every added loss component is net-negative in ablation. (4) **Linear projection guidance**: Procrustes alignment with $\geq$100 parallel pairs is the only consistently beneficial label-free method, but its effect reverses for languages with zero-shot AUROC > 0.62, precluding a single actionable threshold. We release all code and negative results to prevent redundant exploration of closed paths.

1 Introduction

Recent work has demonstrated that large language models encode linear correctness directions in their internal representations **??**. Probes trained on English representations achieve AUROC > 0.75 for predicting answer correctness, enabling representation-level calibration that is complementary to output-based methods. However, all probe-based calibration results to date are English-only, leaving a critical question unanswered: *do these correctness directions transfer across languages?*

This question matters because non-English languages exhibit substantially worse calibration **?**, and the most promising solutions—per-language supervised probes or ensemble methods like LACE—require target-language labels that are expensive for low-resource settings. If English correctness directions transfer, they offer a zero-label calibration path.

A pilot study on Qwen2.5-7B and Llama-3.1-8B (5 languages each) suggested promising patterns: a local Pareto tradeoff at deepest layers, a Simpson's Paradox in CKA–transfer correlations, and a natural distributional matching approach (CaCLR) to test. We scaled this to three production architectures, 20 languages, and systematic ablations.

The results are predominantly negative. The pilot patterns do not generalize. MoE architectures show no Pareto frontier; the only dense model (Gemma 4 31B) shows marginal effects far weaker than the pilot. The Simpson's Paradox in CKA is a small-sample artifact. CaCLR and all distributional matching variants fail. Difficulty–capability decomposition is indistinguishable from random projection.

However, we contribute four findings of lasting value:

1. **Control task methodology warning** (§9): 35% of cross-lingual probe results are capacity artifacts, with implications for the entire probing literature.
2. **Mean-shift impossibility** (§4): a formal proof that mean-shift cannot improve AUROC under standard probe pipelines.
3. **Distributional matching impossibility** (§7): systematic closure of MMD, CORAL, and high-dimensional variants.
4. **Conditional linear projection guidance** (§10): the only viable label-free path, with resource-tier-dependent sample efficiency.

2 Related Work

Probe-based calibration. Burns et al. extract contrast-consistent directions; Miao et al. train correctness probes with cross-task transfer (CORAL); Miao & Ungar demonstrate correctness–confidence orthogonality and probe-based steering. All work exclusively in English.

Cross-lingual probing. Xiao et al. transfer knowledge-boundary probes (input-level answerability) across 8 languages. Critically, answerability is output-independent while correctness depends on the model's actual answer—a fundamentally different transfer problem. Civelli et al. transfer difficulty probes cross-lingually; difficulty serves as our ablation control.

Domain adaptation theory. Ben-David et al. bound target-domain error via source error plus $\mathcal{H}$-divergence. Zhao et al. prove that marginal distribution matching fails when conditional distributions $P(Y|X)$ differ across domains—directly predicting our CaCLR negative result.

Cross-lingual alignment. MUSE and VecMap learn linear projections between embedding spaces. Our Procrustes alignment is conceptually analogous at the sentence-level correctness representation.

Probing methodology. Hewitt & Liang propose control tasks measuring probe selectivity; Voita & Titov use MDL probes. We extend selectivity checks to the cross-lingual setting for the first time.

3 Experimental Setup

3.1 Models

We probe three architectures: **Qwen3-235B-A22B** (94 layers, MoE, INT4 quantized, 24 probed layers every 4th); **Llama 4 Scout** (48 layers, 17B-16E MoE, W4A16, 12 probed layers every 4th); **Gemma 4 31B** (60 layers, dense, BF16 unquantized, 16 probed layers every 4th + final). Gemma serves as a quantization-free dense control.

3.2 Languages and Data

We evaluate 20 typologically diverse languages from GlobalMMLU, grouped into three resource tiers: **HIGH** (7): zh, es, fr, de, ja, pt, it; **MID** (7): ar, hi, ru, ko, tr, vi, th; **LOW** (6): sw, bn, am, yo, te, my. Probes are trained on the English portion (70% stratified split for training, 30% for evaluation). Additional benchmarks (Belebele, MMLU-ProX) are evaluated in Appendix B.

Table 1: Qwen3-235B: per-language depth–transfer Spearman ρ. Only 1/20 languages reaches significance.

Lang	Tier	ZS AUROC	ρ	p	Pattern
zh	HIGH	.668	$-.31$	.14	weak neg
es	HIGH	.675	$-.25$	.24	n.s.
fr	HIGH	.658	$-.28$	.18	n.s.
de	HIGH	.649	$-.19$	.38	n.s.
ja	HIGH	.588	$+.18$	.40	U-shape
pt	HIGH	.662	$-.22$	.30	n.s.
it	HIGH	.645	$-.16$	.46	n.s.
ar	MID	.608	$+.05$	.82	flat
hi	MID	.567	$-.41$	**.046**	sig. neg
ru	MID	.621	$-.14$	.52	n.s.
ko	MID	.591	$+.25$	.24	inverted
tr	MID	.548	$-.08$	.71	n.s.
vi	MID	.562	$+.09$	.68	flat
th	MID	.528	$-.11$	.61	n.s.
sw	LOW	.518	$-.35$	.09	marginal
bn	LOW	.524	$-.12$	.58	n.s.
am	LOW	.486	$+.14$	.52	non-mono
yo	LOW	.475	$-.02$	.93	flat
te	LOW	.502	$-.18$	.40	n.s.
my	LOW	.468	$+.10$	.64	non-mono

3.3 Transfer Conditions

Five conditions are compared: (1) **Zero-shot**: apply English-trained probe directly; (2) **Mean-shift**: subtract mean activation difference between English and target; (3) **Linear projection** (Procrustes, 200 parallel pairs); (4) **Few-shot** (25 or 100 target-language labels); (5) **CaCLR**: joint calibration + distributional matching (details in §7).

3.4 Probes and Metrics

Logistic regression probes (justified in Appendix A); AUROC for discrimination, ECE for calibration. A language "passes" zero-shot transfer if AUROC > 0.58.

4 Mean-Shift Impossibility

Under StandardScaler preprocessing followed by logistic regression, mean-shift adds a constant offset c to all probe logits: $\hat{y}_{\text{shifted}}(x) = w^\top(x + \delta) + b = \hat{y}(x) + w^\top\delta$. Since AUROC depends only on ranking, and a constant offset preserves ranking, $\Delta\text{AUROC} \equiv 0$ *mathematically*.

All 60 model–language combinations confirm $\Delta\text{AUROC} = 0.000$ (Tables **??**, **??**, **??**). This distinguishes our pipeline from Xiao et al., where mean-shift operates under different preprocessing.

5 Pareto Frontier Analysis

The pilot study on 7–8B dense models suggested a depth–transfer Pareto tradeoff. We test this across three full-scale architectures.

Table 2: Dense vs MoE architecture comparison.

Metric	Qwen3 (MoE)	Llama4 (MoE)	Gemma4 (Dense)
EN best AUROC	.741	.718	.755
Global ρ	$-.12$ (n.s.)	$-.06$ (n.s.)	$-.24$ (p=.038)
Per-lang sig.	1/20	0/20	4/20
Non-monotonic	13/20	8/20	5/20
Final drop (pp)	0.6	0.3	0.8
ZS pass rate	55%	45%	60%

Table 3: Transfer conditions averaged across 3 architectures and 20 languages.

Condition	AUROC	ECE	Δ AUROC	Win%
Zero-shot	.574	.282	—	—
Mean-shift	.574	.288	.000	0%
CaCLR (best)	.566	.275	$-.008$	20%
Lin. proj. (200)	.610	.245	$+.036$	70%
Few-shot (25)	.596	.258	$+.022$	52%
Few-shot (100)	.623	.232	$+.049$	75%

5.1 MoE Architectures: No Pareto

Qwen3-235B (Table 1) shows global $\rho = -0.12$ (p=0.078), with 13/20 languages exhibiting non-monotonic depth–transfer patterns. Only Hindi reaches per-language significance ($\rho = -0.41$, p=0.046). Final-layer transfer drop is 0.6pp, far below the pilot's 2–4pp.

Llama 4 Scout shows even weaker effects: global $\rho = -0.06$ (p=0.31), 0/20 per-language significant, 8/20 non-monotonic. Final-layer drop is 0.3pp.

MoE expert routing breaks the depth→specialization monotonic mapping that underlies Pareto in dense models. Different experts are activated at different depths for different languages, creating non-monotonic transfer profiles.

5.2 Dense Architecture: Marginal Pareto

Gemma 4 31B Dense shows the clearest depth–transfer relationship: global $\rho = -0.24$ (p=0.038), 4/20 per-language significant (zh, fr, hi, sw), final-layer drop 0.8pp. But even this is far weaker than the pilot: only 4/20 languages significant, and the 0.8pp final-layer drop is marginal.

Table 2 summarizes: the Pareto tradeoff is architecture-conditional (present in dense, absent in MoE) and far weaker at full scale than the pilot predicted. Three-architecture mean EN AUROC is 0.738.

5.3 Transfer Conditions

Table 3 shows the five-way comparison. Mean-shift is provably inert (§4). CaCLR degrades performance (§7). Linear projection and few-shot are the only beneficial conditions.

6 CKA Analysis: Simpson's Paradox Falsified

The pilot found a Simpson's Paradox: raw CKA–transfer correlation was negative (depth confound), but partial correlation controlling for depth was positive and significant (Llama pilot: $\rho = +0.41$, p=0.006). Full-scale results falsify this.

Table 4: all three partial correlations are non-significant (max $\rho = +0.16$, min p=0.09). The Llama pilot's $\rho = +0.41$ collapses to $+0.06$ with 20 languages, confirming it as a 5-language small-sample artifact. CKA provides no explanatory power beyond depth.

Table 4: Simpson's Paradox validation. Partial correlations controlling for depth are all non-significant.

Model	Raw ρ	Raw p	Partial ρ	Partial p	Verdict
Qwen3	$-.32$	$<.001$	$+.11$	.28	n.s.
Llama4	$-.38$	$<.001$	$+.06$	.45	artifact
Gemma4	$-.29$	$<.001$	$+.16$	.09	marginal

Table 5: Subspace angle between EN and target correctness directions at $\sim$60% depth.

Lang	Tier	Qwen3	Llama4	Gemma4
es	HIGH	15°	20°	14°
zh	HIGH	18°	24°	16°
ja	HIGH	35°	42°	32°
ar	MID	32°	40°	30°
ko	MID	28°	36°	26°
hi	MID	38°	46°	35°
tr	MID	45°	52°	42°
sw	LOW	52°	58°	48°
bn	LOW	48°	55°	45°
am	LOW	62°	65°	58°

Same-layer analysis. Within individual layers, CKA–transfer correlation is weak: median ρ=0.14 across all layers and models, with only 3/52 probed layers reaching significance.

Subspace angle analysis. Correctness direction angles between English and target languages vary dramatically by resource tier: HIGH 15–38°, MID 28–52°, LOW 42–65° (Table 5). Within-language angle standard deviation across layers is 8–15°, indicating the correctness direction is neither language-agnostic nor layer-stable.

7 CaCLR and Distributional Matching

CaCLR optimizes $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{consistency}}$, where $\mathcal{L}_{\text{consistency}}$ uses MMD on probe logits from parallel inputs. We test whether distributional matching can improve cross-lingual transfer.

7.1 Implementation

MMD uses a Gaussian RBF kernel with median-heuristic bandwidth. Optimization: Adam, lr=10^{-3} with cosine schedule to 10^{-5}, 50 epochs, patience-10 early stopping on held-out English BCE loss. Initialized from the standard LR solution. Batch size 256. Parallel set sizes range from 618 to 2,852 per language.

7.2 β Sweep: Consistency Loss is Inert or Harmful

Table 6: every β value harms average performance. Higher β monotonically increases harm, with β=1.0 degrading 75% of language–architecture combinations. Multi-seed evaluation (5 seeds, std = 0.003) confirms these are stable results. A mixed β criterion selecting on held-out zh+sw matches English-only selection (β=0.001), ruling out systematic bias against cross-lingual benefit.

7.3 Distributional Matching: All Paths Closed

We extend beyond 1-D MMD to test whether the failure is MMD-specific or fundamental.

Table 7: MMD fails across dimensionalities (1-D, 50-D, 100-D). Higher dimensions are worse due to curse of dimensionality with $\sim$2000 training samples. CORAL (second-order alignment) also fails.

Table 6: CaCLR β sweep (3 architectures $\times$ 20 languages).

β	AUROC	Δ vs ZS	% Improved	% Harmed
0.001	.572	$-.002$	30%	42%
0.01	.569	$-.005$	25%	52%
0.1	.565	$-.009$	18%	62%
1.0	.558	$-.016$	12%	75%

Table 7: Distributional matching comprehensive ablation. All methods degrade performance. Linear projection (non-distributional) shown for contrast.

Method	Dim.	Δ AUROC
MMD (logit)	1-D	$-.004$
MMD (PCA-50)	50-D	$-.006$
MMD (PCA-100)	100-D	$-.008$
CORAL (logit)	1-D	$-.005$
CORAL (PCA-50)	50-D	$-.007$
Linear projection	full-D	$+.036$

This extends the negative result from "MMD fails" to "distributional matching fails": the cross-lingual correctness calibration problem violates the covariate shift assumption ($P(Y|X)$ differs across languages) that these methods require.

7.4 Component Ablation

Table 8: BCE-only retraining approximately recovers the zero-shot baseline (the LR solution is already optimal). Each additional component degrades performance. This is not a hyperparameter problem—it is a fundamental mismatch between the distributional matching objective and the conditional-distribution structure of cross-lingual correctness.

8 Difficulty Decomposition

We test whether transfer degradation decomposes into a difficulty-shared (language-agnostic) and capability-specific (language-dependent) component. IRT difficulty probes (R^2=0.28) are trained, and the difficulty direction is projected out before retraining correctness probes. A random-direction control projects out a randomly sampled direction of equal norm.

Mean decomposition improvement: +0.2pp (range: -0.8 to +1.2pp). Random-direction control: -0.1pp (range: -1.0 to +0.8pp). Paired test: p=0.38. The decomposition is statistically indistinguishable from projecting out a random direction: observed deltas arise from dimensionality reduction effects, not difficulty-specific signal removal. LOW-tier languages show negative deltas in both conditions—projecting out any direction damages their already-weak signal. Difficulty and capability are entangled in the representation and cannot be operationally separated.

9 Control Task Analysis

We extend Hewitt & Liang's control task framework to cross-lingual probing for the first time. For each language–layer–model combination, we train probes on randomly permuted labels and compute selectivity $= \text{AUROC}_{\text{real}} - \text{AUROC}_{\text{permuted}}$. A combination is flagged as a potential artifact if selectivity < 0.05.

35% overall artifact rate. Table 10: more than a third of cross-lingual probe results fail the selectivity check. LOW-resource languages suffer most (42%), but MID-resource languages are also substantially

Table 8: Component ablation: each added loss is net-negative.

Configuration	Δ vs ZS	Note
BCE only (retrain)	$-.001$	$\approx$ ZS
BCE + SB-ECE	$-.005$	ECE hurts discrim.
BCE + MMD (1-D)	$-.004$	inert
BCE + MMD (PCA-50)	$-.006$	fails
BCE + MMD (PCA-100)	$-.008$	worse
BCE + CORAL	$-.005$	fails
BCE + CORAL (PCA-50)	$-.007$	fails
Full CaCLR	$-.008$	compound degrad.

Table 9: Difficulty decomposition vs random-direction control (3-model mean). Paired p-values test decomposition vs random.

Lang	Tier	Orig. ZS	After proj.	Δ (pp)	Rand Δ
zh	HIGH	.666	.670	+0.4	+0.1
es	HIGH	.674	.676	+0.2	-0.1
fr	HIGH	.657	.660	+0.3	+0.2
de	HIGH	.650	.652	+0.2	-0.2
hi	MID	.563	.565	+0.2	+0.3
ru	MID	.621	.622	+0.1	-0.2
ko	MID	.589	.598	+0.9	+0.2
tr	MID	.546	.540	-0.6	-0.4
sw	LOW	.515	.510	-0.5	-0.3
bn	LOW	.521	.519	-0.2	$+0.0$

affected (31%).

Per-language selectivity. Table 11 shows per-language results for Qwen3. Four LOW-resource languages (am, yo, my, sw) have selectivity < 0.01 at the English-optimal layer—their probes learn capacity patterns, not correctness signal. Three MID-resource languages (tr: 0.031, th: 0.028, vi: 0.042) also fall below or near the threshold.

Selectivity predicts transfer. Selectivity correlates with zero-shot AUROC at ρ=0.65, suggesting that published positive cross-lingual probing results may partially reflect capacity artifacts rather than genuine correctness signal transfer.

Threshold sensitivity. At threshold 0.03, overall artifact rate rises to 42% (LOW: 52%); at 0.10, it drops to 24% (LOW: 35%). The finding is robust across thresholds.

10 Linear Projection Analysis

Linear projection (Procrustes alignment) is the only consistently positive label-free method (+3.6pp vs zero-shot, Table 3). We analyze its sample efficiency and limitations.

10.1 Sample Efficiency

Table 12: 100 pairs achieve 96% of the 200-pair performance, but the pilot's claim that 50 pairs suffice is overly optimistic. Saturation varies by tier: HIGH languages reach 95% at $\sim$62 pairs (range 38–92), MID at $\sim$118 (72–158), LOW at $\sim$175 (140–198). Low-resource languages require more pairs because their

Table 10: Cross-lingual artifact rates by resource tier (3 models, selectivity threshold 0.05).

Tier	n langs	Artifact rate	Sel. mean	Sel. std
HIGH	7	15%	.195	.042
MID	7	31%	.138	.038
LOW	6	42%	.082	.035
ALL	20	35%	.142	.058

Table 11: Per-language selectivity at EN-optimal layer (Qwen3-235B). "†" indicates selectivity < 0.05.

Lang	Tier	Real	Perm.	Sel.	Flag
en	—	.741	.488	.253	
zh	H	.668	.495	.173	
es	H	.675	.492	.183	
fr	H	.658	.490	.168	
de	H	.649	.486	.163	
ja	H	.588	.501	.087	
pt	H	.662	.489	.173	
it	H	.645	.492	.153	
ar	M	.608	.496	.112	
hi	M	.567	.503	.064	
ru	M	.621	.493	.128	
ko	M	.591	.499	.092	
tr	M	.548	.517	.031	†
vi	M	.562	.520	.042	†
th	M	.528	.500	.028	†
sw	L	.518	.510	.008	†
bn	L	.524	.505	.019	†
am	L	.486	.482	.004	†
yo	L	.475	.470	.005	†
te	L	.502	.492	.010	†
my	L	.468	.462	.006	†

larger subspace angles (Table 5) and noisier translations demand more data to stabilize the Procrustes rotation.

10.2 Effect Reversal

Table 13: below ZS AUROC $\sim$0.53, projection consistently helps; above $\sim$0.62, it consistently hurts. The inconsistent zone (0.53–0.62) varies $\pm$0.08 across languages—Japanese (HIGH, 0.588) is helped while Korean (MID, 0.589) is hurt at nearly identical baseline AUROCs. No single threshold is reliable. Practical recommendation: apply projection when ZS AUROC < 0.50 (likely beneficial), avoid when $>$ 0.65 (likely harmful), and validate on held-out data otherwise.

11 Supporting Results

11.1 Cross-Lingual Orthogonality

Correctness–confidence orthogonality (cosine similarity) persists for HIGH-resource languages (mean cosine $= 0.035$) but breaks down for LOW-resource (mean $= 0.108$, std $= 0.078$). Architecture agreement is 65% (13/20 languages). The elevated LOW-resource cosine likely reflects probe noise rather than true non-orthogonality, consistent with the high artifact rates in §9.

Table 12: Linear projection sample efficiency (3 architectures $\times$ 20 languages).

n pairs	AUROC	% of 200	Std	Sig. %
10	.552	66%	.062	22%
20	.571	78%	.055	38%
50	.592	89%	.049	52%
100	.605	96%	.044	65%
200	.610	100%	.042	70%

Table 13: Procrustes effect by language (200 pairs). Effect reverses above ZS AUROC $\sim$0.62.

Lang	Tier	ZS	LinProj	Δ	Effect
am	LOW	.483	.508	$+.025$	helps
yo	LOW	.468	.494	$+.026$	helps
sw	LOW	.511	.528	$+.017$	helps
tr	MID	.544	.560	$+.016$	helps
hi	MID	.560	.568	$+.008$	marginal
vi	MID	.563	.562	$-.001$	neutral
ko	MID	.589	.582	$-.007$	hurts
ar	MID	.606	.598	$-.008$	hurts
ru	MID	.622	.610	$-.012$	hurts
es	HIGH	.674	.654	$-.020$	hurts
ja	HIGH	.588	.596	$+.008$	helps

12 Discussion

Why do pilot results not generalize? The pilot used 7–8B dense models with 5 languages. Dense architectures have monotonic depth→specialization mappings; MoE routing breaks this. Five languages over-represent HIGH-resource cases where effects are strongest. Both the Pareto finding and the Simpson's Paradox were small-sample artifacts amplified by architectural specificity.

Why does distributional matching fail? Zhao et al. predict that marginal distribution matching fails when $P(Y|X)$ differs across domains. Cross-lingual correctness probing violates covariate shift: $P(\text{correct}|\text{representation})$ differs systematically across languages because model capability varies. MMD matches marginals but cannot align the conditional structure that carries correctness information. This is confirmed by the monotonic degradation with β (Table 6) and the failure of both CORAL and high-dimensional MMD (Table 7).

Field implications of control task findings. The 35% artifact rate implies that cross-lingual probing studies—including those reporting only MID-resource languages—should routinely include selectivity checks. The ρ=0.65 selectivity–AUROC correlation suggests that effect sizes in the literature may be inflated by capacity artifacts.

Limitations. (1) MCQA-only evaluation; correctness probing may behave differently for open-ended generation. (2) Quantization artifacts at deep layers introduce 1–2pp bias for Qwen3 (INT4) and Llama4 (W4A16); Gemma4 (BF16) is unaffected (Appendix C). (3) Linear projection requires parallel data; availability varies by language.

13 Conclusion

We present the first systematic study of cross-lingual correctness probe transfer. Our results close several paths: distributional matching (MMD, CORAL, high-dimensional variants) consistently harms transfer;

mean-shift is provably inert; the Pareto tradeoff is architecture-dependent and marginal at scale; CKA provides no explanatory power beyond depth; difficulty–capability decomposition is not operational.

Two positive contributions remain. First, control task analysis reveals that 35% of cross-lingual probe results are capacity artifacts—a methodological warning for the field. Second, linear projection with $\geq$100 parallel pairs is the only consistently beneficial label-free method, though its effect reverses for high-performing languages. These findings prevent the community from pursuing demonstrably closed paths while providing conditional practical guidance.

A Probe Complexity Comparison

Table 14: Logistic regression vs 2-layer MLP (3 models $\times$ 20 languages).

Metric	LR	MLP	Δ	p
EN AUROC	.738	.752	+.014	.006
ZS degradation	15.8%	19.4%	+3.6pp	.018
ZS AUROC	.621	.608	$-.013$	.032
EN ECE	.238	.252	+.014	.058
LR better for xfer	72%	28%	—	—

MLP gains +1.4pp EN AUROC but transfers worse ($-$3.6pp more degradation). LR is preferred for cross-lingual transfer in 72% of languages.

B Cross-Benchmark Generalization

Table 15: Cross-benchmark generalization.

Benchmark	ZS pass	ZS AUROC	CaCLR Δ	LinProj Δ	Corr.
GlobalMMLU	57%	.574	$-.008$	$+.036$	—
Belebele	45%	.545	$-.015$	$+.028$	.71
MMLU-ProX	55%	.565	$-.009$	$+.033$	.79

Cross-benchmark correlation is moderate (.71–.79). Belebele (reading comprehension) shows lower transfer, suggesting correctness probing is partially MCQA-format-specific. CaCLR negative results replicate across benchmarks.

C Quantization Validation

Table 16: Quantization impact on probe AUROC (5 representative layers).

Model	Quant	Depth%	FP16	Quant	Δ
Llama4	W4A16	20%	.650	.647	$-.003$
Llama4	W4A16	60%	.698	.688	$-.010$
Llama4	W4A16	95%	.715	.699	$-.016$
Qwen3	INT4	20%	.628	.623	$-.005$
Qwen3	INT4	60%	.705	.692	$-.013$
Qwen3	INT4	95%	.741	.720	$-.021$

Quantization introduces 0.3–2.1pp AUROC degradation, increasing with depth. Rank preservation: Llama4 92%, Qwen3 89%. Deep-layer conclusions are potentially affected, but the negative results

(distributional matching failure, decomposition null) do not depend on precise values. Gemma4 (BF16) provides quantization-free validation.

D Extended Language Coverage

Extending from 20 to 52 languages: ZS pass rate drops from 57% to 50% (more LOW-resource languages added). LOW-tier artifact rate rises to 45%. Core conclusions are unchanged; expanded coverage strengthens the control task finding.